



SARVEPALLI RADHAKRISHNAN UNIVERSITY, BHOPAL

BE 1st SEMESTER

BE 11 PHYSICS

TOTAL -60 HOURS

OBJECTIVE:

The students are to be learn the below mentioned:

- Electrostatic in vacuum
- Electrostatic in linear dielectric medium
- Magnetostatic in linear magnetic medium
- Faraday's law
- Electromagnetic waves

Module 1: Wave nature of particles and the Schrodinger equation (12 HOURS)

Introduction to Quantum mechanics, Wave nature of Particles, operators, Time-dependent and time independent Schrodinger equation for wave function, Application: Particle in a one dimensional Box, Born interpretation, Free-particle wave function and wave-packets, v_g and v_p relation Uncertainty principle. Module

2: Wave optics ((12 HOURS)

Huygens' principle, superposition of waves and interference of light by wave front splitting and amplitude splitting: Young double slit experiment, Newton's rings, Michelson interferometer, MachZehnder interferometer, Fraunhofer diffraction from a single slit and a circular aperture, the Rayleigh criterion for limit of resolution and its application to vision; Diffraction gratings and their resolving power. Module

3: Introduction to solids ((12 HOURS)

Free electron theory of metals, Fermi level of Intrinsic and extrinsic, density of states, Bloch's theorem for particles in a periodic potential, Kronig-Penney model (no derivation) and origin of energy bands. V-I characteristics of PN junction, Zener diode, Solar Cell, Hall Effect.

Module 4: Lasers ((12 HOURS)

Einstein's theory of matter radiation interaction and A and B coefficients; amplification of light by population inversion, different types of lasers; gas lasers (He-Ne, CO₂), solid-state lasers (ruby, Neodymium), Properties of laser beams: mono-chromaticity, coherence, directionality and brightness, laser speckles, applications of lasers in science, engineering and medicine. Introduction to Optical fiber, acceptance angle and cone, Numerical aperture, V number, attenuation. Module

5: Electrostatics in vacuum (12 HOURS)

Calculation of electric field and electrostatic potential for a charge distribution; Electric



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displacement, Basic Introduction to Dielectrics, Gradient, Divergence and curl, Stokes' theorem, Gauss Theorem, Continuity equation for current densities; Maxwell's equation in vacuum and non-conducting medium, Poynting vector.

List of Experiment

1. To determine the dispersive power of prism.
2. To determine the λ of sodium light with the help of newton's Ring.
3. Resolving Power of Telescope.
4. YDSE (Young's double slit Experiment).
5. To determine the frequency of AC mains supply.
6. V-I Characteristics of P-N junction diode.
7. To determine the λ of diode lines by single slit diffraction.
8. To determine the plank's constant with the help of photocell.
9. Hall's effect experiment.
10. Calibration of ammeter by using reference zener diode. w.e.f. July 2018
11. To study the effect of temperature on reverse saturation current in P-N junction diode and to determine the energy band gap.
12. To determine the λ of sodium by using plane diffraction grating.
13. To determine the prominent lines of mercury source by plane diffraction grating.
14. To determine the numerical aperture of an optical fiber.
15. To determine λ of given laser by plane diffraction grating.

OUTCOME:

- Experimental exposure to electromagnetic induction and electromagnetic braking.
- LC circuit and LCR circuit
- Resonance phenomenon in LCR circuit
- Magnetic field from Helmholtz coil
- Measurement of Lorentz force in a vacuum tube

Suggested Reference Books

1. A. Ghatak, Optics.
2. O. Svelto, Principles of Lasers.
3. David Griffiths, Introduction to Electrodynamics.
4. D.J. Griffiths, Quantum Mechanics.
5. Halliday & Resnick, Physics.



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BE 1st SEMESTER BE 12 MATHEMATICS-I

TOTAL -60 HOURS

OBJECTIVE:

- The objective of their course is to familiarize the prospective engineers with techniques in calculus, multivariate analyzer and linear algebra. it aim to equip the students with standard concepts and tools at an intermediate to advance level that will serve them well towards taking more advanced level of mathematics and applications that they would find useful in their disciplines.

Module 1: Calculus:

Evolutes and involutes; Evaluation of definite and improper integrals; Beta and Gamma functions and their properties; Applications of definite integrals to evaluate surface areas and volumes of revolutions. (12 HOURS)

Module 2: Calculus:

Rolle's Theorem, Mean value theorems, Taylor's and Maclaurin theorems with remainders; indeterminate forms and L'Hospital's rule; Maxima and minima. (12 HOURS)

Module 3: Sequences and series:

Convergence of sequence and series, tests for convergence; Power series, Taylor's series, series for exponential, trigonometric and logarithm functions; Fourier series; Half range sine and cosine series, Parseval's theorem. (12 HOURS)

Module 4: Multivariable Calculus (Differentiation):

Limit, continuity and partial derivatives, directional derivatives, total derivative; Tangent plane and normal line; Maxima, minima and saddle points; Method of Lagrange multipliers; Gradient, curl and divergence. (12 HOURS)

Module 5: Matrices

Inverse and rank of a matrix, rank-nullity theorem; System of linear equations; Symmetric, skewsymmetric and orthogonal matrices; Determinants; Eigenvalues and eigenvectors; Diagonalization of matrices; Cayley-Hamilton Theorem, and Orthogonal transformation.

(12 HOURS)

OUTCOME:

The student will learn

- To apply differential and integral calculus to notions of curvature and to improper integrals, apart from some other applications they will have a basic understanding of beta and gamma functions.
- The fallout of Rolle's Theorem that is fundamental to application of analysis to engineering problems.
- The tool of power series and Fourier series for learning advanced Engineering Mathematics.



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- To deal with functions of several variables those are essential in most branches of engineering.
- The essential tool of matrices and linear algebra in comprehensive manner.

BOOKS:

1. G.B. Thomas and R.L. Finney, Calculus and Analytic geometry, 9th Edition, Pearson, Reprint, 2002.
2. Erwin kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006.
3. Vecrarajan T., Engineering Mathematics for first year, Tata McGraw-Hill, New Delhi, 2008.
4. Ramana B.V., Higher Engineering Mathematics, Tata McGraw Hill New Delhi, 11th Reprint, 2010.
5. D. Poole, Linear Algebra: A Modern Introduction, 2nd Edition, Brooks/Cole, 2005.
6. N.P. Bali and Manish Goyal, A text book of Engineering Mathematics, Laxmi Publications Reprint, 2008.
7. B.S. Rewal, Higher Engineering Mathematics, Kanna Publishers, 36th Edition, 2010.



SARVEPALLI RADHAKRISHNAN UNIVERSITY, BHOPAL

BE 1st SEMESTER

BE 13 BASIC ELECTRICAL & ELECTRONICS ENGINEERING

TOTAL -60 HOURS

OBJECTIVE

- To understand and analysis basic electrical and magnetic circuits
- To study the working principle of electrical machine and power converters
- To introduce the components of low voltage electrical installations.

MODULE I

Electrical circuit analysis- Voltage and current sources, dependent and independent sources, source conversion, DC circuits analysis using mesh & nodal method, Thevenin's & superposition theorem, star-delta transformation, phase AC circuits under sinusoidal steady state, active, reactive and apparent power, physical meaning of reactive power, power factor, 3-phase balanced and unbalanced supply, star and delta connections. **(12 HOURS)**

Module II

Transformers- Review of laws of electromagnetism, mmf, flux, and their relation, analysis of magnetic circuits, Single-phase transformer, basic concepts and construction features, voltage, current and impedance transformation, equivalent circuits, phasor diagram, voltage regulation, losses and efficiency, OC and SC test. **(12 HOURS)**

Module III

Rotating Electric machines- Constructional details of DC machine, induction machine, and synchronous machine, Working principle of 3-Phase induction motor, Emf equation of 3-Phase induction motor, Concept of slip in 3-Phase induction motor, Explanation of Torque-slip characteristics of 3-Phase induction motor, Classification of self excited DC motor and generator.

(12 HOURS)

Module IV

Digital Electronics- Number systems used in digital electronics, decimal, binary, octal, hexadecimal, their complements, operation and conversion, floating point and signed numbers, Demorgan's theorem, AND, OR, NOT, NOR, NAND, EX-NOR, EX-OR gates and their representation, truth table, half and full adder circuits, R-S flip flop, J-K flip flop. **(12 HOURS)**

Module V

ELECTRONIC COMPONENTS AND CIRCUITS- Introduction to Semiconductors, Diodes, V-I characteristics, Bipolar junction transistors (BJT) and their working, introduction to CC, CB & CE transistor configurations, different configurations and modes of operation of BJT, DC biasing of BJT.

(12 HOURS)



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GUT COME:

- Get an exposure to common electrical component and their ratings
- Make electrical connections by wire of appropriate ratings.
- Understand the usages of common electrical measuring instruments.
- Understand the basic characteristics of transformer and electrical machines.
- Get an exposure to the working of power electronic converters.

BOOKS:

1. Vincent Del Toro, Electrical Engineering Fundamentals, PHI Learning, II Edition
2. S.Ghosh, Fundamentals of Electrical and Electronics Engineering, PHI, II Edition.
3. Millman, Halkias & Parikh, Integrated Electronics, Mc Graw Hill, II Edition
4. Nagrath & Kothari, Basic Electrical Engineering, III Edition TMH.
5. J.S. Katre, Basic Electronics Engg, Max Pub. Pune.
6. Hughes, Electrical and Electronic Technology, Pearson Education IX Edition



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BE 1st SEMESTER

BE 14 ENGINEERING GRAPHICS

TOTAL -60 HOURS

OBJECTIVE:

- All phase of manufacturing or construction requires the conversion of new ideas and design concept into the basic line language of graphics.

Module I

Scales: Representative factor, plain scales, diagonal scales, scale of chords.

Conic sections: Construction of ellipse, parabola, hyperbola by different methods; Normal and Tangent.

Special Curves: Cycloid, Epi-cycloid, Hypo-cycloid, Involute, Archimedean and logarithmic spirals.

(10 HOURS)

Module II

Projection: Types of projection, orthographic projection, first and third angle projection, **Projection of points and lines**, Line inclined to one plane, inclined with both the plane, True Length and True Inclination, Traces of straight lines. (10 HOURS)

Module III

Projection of planes and solids: Projection of Planes like circle and polygons in different positions; Projection of polyhedrons like prisms, pyramids and solids of revolutions like cylinder, cones in different positions. (10 HOURS)

Module IV

Section of Solids: Section of right solids by normal and inclined planes; Intersection of cylinders.
Development of Surfaces: Parallel line and radial - line method for right solids. (10 HOURS)

Module V

Isometric Projections: Isometric scale, Isometric axes, Isometric Projection from orthographic drawing. (10 HOURS)

Module VI

Computer Aided Drafting (CAD): Introduction, benefit, software's basic commands of drafting entities like line, circle, polygon, polyhedron, cylinders; transformations and editing commands like move, rotate, mirror, array; solution of projection problems on CAD. (10 HOURS)



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OUTCOME:

The student becomes aware of below:

- Introduction to Engineering drawing covering
- Orthographic projections
- Section of solids
- Isometric projection
- Computer aided drafting

BOOKS:

1. Visvesvaraya Tech. University; A Premier on Computer Aided Engg drawing; VTU Belgaum
2. Bhatt N.D.; Engineering Drawing, Charotar
3. Venugopal K.; Engineering Graphics; New Age
4. John KC; Engg. Graphics for Degree; PHI.
5. Gill P.S.; Engineering Drawing; kataria
6. Jeyopooan T.; Engineering drawing & Graphics Using AutoCAD; Vikas
7. Agrawal and Agrawal; Engineering Drawing; TMH
8. Shrivastava MB and Rana BC; Engg. drawing; Pearson Education
9. Luzadder WJ and Duff JM; Fundamental of Engg Drawing; PHI
10. Jolhe DA Engg. Drawing an Introduction; TMH
11. Narayana K.L.; Engineering Drawing; Scitech



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BE 1st SEMESTER

BE 15 BASIC CIVIL ENGINEERING & ENGINEERING MECHANICS

TOTAL -60 HOURS

OBJECTIVE:

- Resolve the forces.
- Find the resultant of given force system
- Find the reaction of beam
- Find the centre of gravity of composite solid.
- Find M.A., V.R., Efficiency and establish law of machine

Module I Building Materials & Construction :

Stones, bricks, cement, lime, timber-types, properties, test & uses, laboratory tests concrete and mortar Materials: Workability, Strength properties of Concrete, Nominal proportion of Concrete preparation of concrete, compaction, curing. Elements of Building Construction, Foundations conventional spread footings, RCC footings, brick masonry walls, plastering and pointing, floors, roofs, Doors, windows, lintels, stair cases types and their suitability. (12 HOURS)

Module II Surveying & Positioning:

Introduction to surveying Instruments – levels, theodolites, plane tables and related devices. Electronic surveying instruments etc. Measurement of distances – conventional and EDM methods, measurement of directions by different methods, measurement of elevations by different methods. Reciprocal leveling. (12 HOURS)

Module –III Mapping & Sensing:

Mapping details and contouring, Profile Cross sectioning and measurement of areas, volumes, application of measurements in quantity computations, Survey stations, Introduction of remote sensing and its applications. (12 HOURS)

Module IV Forces and Equilibrium:

Graphical and Analytical Treatment of Concurrent and non- concurrent Co- planner forces, free Diagram, Force Diagram and Bow's notations; Application of Equilibrium Concepts: Analysis of plane Trusses; Method of joints, Method of Sections. Frictional force in equilibrium problems

(12 HOURS)

Module – V Centre of Gravity and moment of Inertia:

Centroid and Centre of Gravity, Moment Inertia of Area and Mass, Radius of Gyration, Introduction to product of Inertia and Principle Axes. Support Reactions, Shear force and bending moment Diagram for Cantilever & simply supported beam with concentrated, distributed load and Couple. (12 HOURS)



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OUTCOME:

The students understand the below mention:

- Verification of law of triangle law of forces and Lami's theorem.
- Verification of law of parallelogram law of forces.
- Determination of moment of inertia of flywheel.
- Determination of coefficient of friction between two given materials.
- Determination of forces in the member of jib crane

BOOKS:

1. S. Ramamrutam & R.Narayanan; Basic Civil Engineering, Dhanpat Rai Pub.
2. Prasad I.B., Applied Mechanics, Khanna Publication.
3. Punmia, B.C., Surveying, Standard book depot.
4. Shesha Prakash and Mogaveer; Elements of Civil Engg & Engg. Mechanics; PHI
5. S.P, Timoshenko, Mechanics of structure, East West press Pvt.Ltd.
6. Surveying by Duggal – Tata McGraw Hill New Delhi.
7. Building Construction by S.C. Rangwala- Charotar publications House, Anand.
8. Building Construction by Grucharan Singh- Standard Book House, New Delhi
9. Global Positioning System Principles and application- Gopi, TMH
10. R.C. Hibbler – Engineering Mechanics: Statics & Dynamics.
11. A. Borelli & Schmidt- Engineering Mechanics- statics dynamics, Thomson' Books
12. R.K. Rajput, Engineering Mechanics S.Chand & Co.



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BE Ist SEMESTER BE 16 INTRODUCTION TO YOGA

TOTAL -60 HOURS

OBJECTIVE:

The students to be made aware of yoga, concept of kriya yoga, bandha and mudras, yoga vashitha, nadi and kanda.

MODULE-1

- Meaning of yoga, Definition, Aim and Objective of yoga.
- Concept of ideal Hut and ideal place and its relevance in modern times.
- Do's and Don't for a yoga practitioner.
- Goal of Hatha yoga and relevance of hatha yoga in contemporary times. (12 HOURS)

MODULE -2

- Concept of kriya yoga.
- Concept of Asanas (Definition, various classification, number of Asanas).
- Concept of pranayama (Definition and types).
- Concept of shhat karma (Dhoti, Neti, Vasti, Kapalbhathi, Tratak, Noli, and Agnisaar). (12 HOURS)

MODULE -3

- Concept, definition of bandha and mudras.
- Concept, definition, benefits and techniques of pratyahar (Patanjalean and Hathyogic).
- Concept of Dharna and dhyana (Patanjali and non Patanjali Concept). (12 HOURS)

MODULE -4

- Highlights of yoga vashitha.
- Yamas, Niyamas, Asanas, Pranayam, Pratyahar, Dharna And Samadhi According to vashitha sanhita.
- Concept of vayu-prana 5 principal pranas function, location, 5 Uppranas sub ordinate function, location.
- Concept of Nadi and Kanda their location and range, its importance in yoga. (12 HOURS)

MODULE -5

- Raja yoga.
- Gyan Yoga.
- Karma Yoga.
- Bhakti Yoga.
- Mantra Yoga.
- Importance of Haths yoga in daily life and social life. (12 HOURS)



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PRACTICAL

ASANA

1. Halasana
2. Pashchimoutanasana
3. Ardha Matsyendra Asana
4. Akarnadhanurasana
5. Uttan Kurmasana
6. Trikorasana
7. Matsyendrasana
8. Kukkutasana
9. Naukasana
10. Garulasana
11. Karnaperasana
12. Padangusthasana
13. Vatayanasana
14. Guptasana
15. Ugrasana
16. Vipreetkarni
17. Tadaasana
18. Sarvangasana
19. Varkshasana

PRANAYAMA

- | | |
|-------------------|--------------|
| 1. Anulom – vilom | 4. Shitali |
| 2. Sitkari | 5. Bhastrika |
| 3. Ujjayi | |

MUDRA

- | | |
|---------------|----------------|
| 1. Yoga Mudra | 2. Brahm Mudra |
|---------------|----------------|

BHANDHA

- | | |
|----------------------|-------------------|
| 1. Jalandhar Bhandha | 2. Udiyan Bhandha |
|----------------------|-------------------|

MEDITATION

1. 10-15 minutes

OUTCOME:

The students having learned types of asana, pranayama, mudra, bhandha and Meditation.



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BE II nd SEMESTER

BE 21 CHEMISTRY-I

TOTAL -60 HOURS

OBJECTIVE:

The concept developed in their course will aid in qualification of several concepts in chemistry that have been introduced at 10+0 levels in schools. Technology is being increasingly based on the electronic, atomic and molecular level modifications. Quantum theory is more than 100 years old and to understand phenomenon at nanometer levels, one has to be base the description of all chemical processes at molecular levels.

Course Contents:

MODULE -I

Water – Analysis, Treatments and Industrial Applications

Sources, Impurities, Hardness & its units, Determination of hardness by EDTA method, Alkalinity & It's determination and related numerical problems. (9 hours)

MODULE -II

Boiler problem & softening methods

Boiler troubles (Sludge & Scale, Priming & Foaming, Boiler Corrosion, Caustic Embrittlement), Softening methods (Lime-Soda, Zeolite and Ion Exchange Methods) and related numerical problems. (9 hours)

MODULE -III

Lubricants and Lubrication

Introduction, Mechanism of lubrication, Classification of lubricants, significance & determination of Viscosity and Viscosity Index, Flash & Fire Points, Cloud & Pour Points, Aniline Point, Acid Number, Saponification Number, Steam Emulsification Number and related numerical problems. (9 hours)

MODULE -IV

Polymer & polymerization

Introduction, types of polymerisation, Classification, mechanism of polymerisation (Free radical & Ionic polymerization). Thermoplastic & Thermosetting polymers Elementary idea of Biodegradable polymers, preparation, properties & uses of the following polymers- PVC, PMMA, Teflon, Nylon 6, Nylon 6:6, Polyester phenol formaldehyde, Urea- Formaldehyde, Buna N, Buna S, Vulcanization of Rubber. (9 hours)

MODULE -V

Phase equilibrium and Corrosion

Phase diagram of single component system (Water) Phase diagram of binary Eutectic System (Cu-Ag.) Corrosion: Types, Mechanisms & prevention. (9 hours)

MODULE -VI

Spectroscopic techniques and application



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Ammonium Sulphate & Copper Sulphate.

Reference Books :

- 1 Chemistry in Engineering and Technology - Vol.1 &2 Kuriacose and Rajaram , McGraw Hill Education
- 2 Fundamental of Molecular Spectroscopy C.N. Banwell , McGraw Hill Education
- 3 Engineering Chemistry – B.K. Sharma, Krishna Prakashan Media (P) Ltd., Meerut.
- 4 Basics of Engineering Chemistry – S.S. Dara & A.K. Singh, S. Chand & Company Ltd., Delhi.
- 5 Applied Chemistry – Theory and Practice, O.P. Viramani, A.K. Narula, New Age International Pvt. Ltd. Publishers, New Delhi.
- 6 Elementary Spectroscopy , Y.R. Sharma , S. Chand Publishing
- 7 Polymer Science, Vasant R. Gowariker, N. V. Viswanathan, Jayadev Sreedhar, New Age International Pvt. Ltd
- 8 Advanced Inorganic Chemistry, G.R. Chatwal, Goal Publishing House
- 9 Engineering Chemistry (NPTEL Web-book) B.L. Tembe, Kamaluddin and M.S. Krishna



SARVEPALLI RADHAKRISHNAN UNIVERSITY, BHOPAL

BE II nd SEMESTER BE 22 MATHEMATICS-II

OBJECTIVE:

TOTAL -60 HOURS

- The objective of their course is to familiarize the prospective engineers with techniques in multivariable integration ordinary and partial differential equations and complex variables. It aims to equip the students to deal with advance level of mathematics and applications that would be essential for their disciplines.

Module 1

Ordinary Differential Equations I : (6 hours) : Differential Equations of First Order and First Degree (Leibnitz linear, Bernoulli's, Exact), Differential Equations of First Order and Higher Degree, Higher order differential equations with constant coefficients, Homogeneous Linear Differential equations, Simultaneous Differential Equations. (12 HOURS)

Module 2

Ordinary differential Equations II: (8 hours) : Second order linear differential equations with variable coefficients, Method of variation of parameters, Power series solutions; Legendre polynomials, Bessel functions of the first kind and their properties. (12 HOURS)

Module 3:

Partial Differential Equations : (8 hours) : Formulation of Partial Differential equations, Linear and Non-Linear Partial Differential Equations, Homogeneous Linear Partial Differential Equations with Constant Coefficients. (12 HOURS)

Module 4

Functions of Complex Variable : (8 hours) : Functions of Complex Variables: Analytic Functions, Harmonic Conjugate, Cauchy-Riemann Equations (without proof), Line Integral, Cauchy-Goursat theorem (without proof), Cauchy Integral formula (without proof), Singular Points, Poles & Residues, Residue Theorem, Application of Residues theorem for Evaluation of Real Integral (Unit Circle).

(12 HOURS)

Module 5

Vector Calculus : (10 hours) : Differentiation of Vectors, Scalar and vector point function, Gradient, Geometrical meaning of gradient, Directional Derivative, Divergence and Curl, Line Integral, Surface Integral and Volume Integral, Gauss Divergence, Stokes and Green theorems. (12 HOURS)

OUTCOME:

The student will learn

- The mathematical tools needed in evaluating multiple integrals and their usages.
- The effective mathematical tools for the solutions of differential equations that model physical



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processes.

- The tools of differentiation and integration of functions of a complex variable that are use various techniques dealing engineering problem.

Textbooks/References:

1. G.B. Thomas and R.L. Finney, Calculus and Analytic geometry, 9th Edition, Pearson, Reprint, 2002.
2. Erwin kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006.
3. W. E. Boyce and R. C. DiPrima, Elementary Differential Equations and Boundary Value Problems, 9th Edn., Wiley India, 2009.
4. S. L. Ross, Differential Equations, 3rd Ed., Wiley India, 1984.
5. E. A. Coddington, An Introduction to Ordinary Differential Equations, Prentice Hall India, 1995.
6. E. L. Ince, Ordinary Differential Equations, Dover Publications, 1958.
7. J. W. Brown and R. V. Churchill, Complex Variables and Applications, 7th Ed., McGraw Hill, 2004.
8. N.P. Bali and Manish Goyal, A text book of Engineering Mathematics, Laxmi Publications, Reprint, 2008.
9. B.S. Grewal, Higher Engineering Mathematics, Khanna Publishers, 36th Edition, 2010.



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BE II nd SEMESTER

BE 23 PROGRAMMING FOR PROBLEM SOLVING WITH C++

TOTAL -60 HOURS

OBJECTIVE :

- To formulate simple algorithms for erythematic and logical problem
- To translate the algorithms to programs (in c++ language)
- To test and execute the programs and correct syntax and logical error.
- To implement conditional branching, iteration recursion.
- To decompose a problem into functions and synthesize a complete program using divide and Conquer approach.
- To use arrays, pointer and structures to formulate algorithms and program.
- To apply programming to solve matrix addition and multiplication problems and searching and sorting problem.

Module –I

Introduction To 'C' Language

Character set, variables and identifiers, built-in data types, variable definition, arithmetic operators and expressions, constants and literals, simple assignment statement, basic input/output statement, simple 'c' programs. (09 HOURS)

Module –II

Conditional Statements And Loops

Decision making within a program, conditions, relational operators, logical connectives, if statement, if-else statement, loops: while loop, do while, for loop, nested loops, infinite loops, switch statement, structured programming. (09 HOURS)

Module –III Arrays

One dimensional arrays: array manipulation; searching, insertion, deletion of an element from an array; finding the largest/smallest element in an array; two dimensional arrays, addition/multiplication of two matrices, transpose of a square matrix; null terminated strings as array of characters, representation sparse matrices. (09 HOURS)

Module –IV

Functions

Top-down approach of problem solving, modular programming and functions, standard library of c



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functions, prototype of a function: formal parameter list, return type, function call, block structure, passing arguments to a function: call by reference, call by value, recursive functions, arrays as function arguments. **(09 HOURS)**

Module –V

Structures and Unions

Structure variables, initialization, structure assignment, nested structure, structures and functions, structures and arrays: arrays of structures, structures containing arrays, unions. **(08 HOURS)**

Module –VI

Pointers

Address operators, pointer type declaration, pointer assignment, pointer initialization, pointer arithmetic, functions and pointers, arrays and pointers, pointer arrays. **(08 HOURS)**

Module –VII File Processing

Concept of files, file opening in various modes and closing of a file, reading from a file, writing onto a file. **(08 HOURS)**

OUTCOME:

- To formulate the algorithms for simple problems.
- To translate given algorithms to a working and correct program.
- To be able to correct syntax errors or reported by the compiler.
- To be able to identify and correct logical error encountered at run time.
- To be able to write iterative as well as recursive programs.

Reference

1. Let Us C-Yaswant kanetkar, BPP
2. Programming in ANSI C-E. Balagurusamy, TMH



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BE II nd SEMESTER BE 24 ENGLISH

TOTAL -60 HOURS

OBJECTIVE:

The student will acquire basic proficiency in English including reading and listening comprehension, writing and speaking skill.

Module I - Languages and skills of communication

Linguistic techniques, Modern usages, Reading comprehension, English phonetic symbols/sounds, Oral presentation, Audition Communication, Processes of Communication, Verbal and Non Verbal Communication, Barriers to Communication. (12 HOURS)

Module II - Application of linguistic ability

Writing of definitions of Engineering terms, Objects, Processes and Principles (Listening) Topics of General Interest, Reproduction from business, daily life, travel, health, buying and selling, company structure, systems etc. (12 HOURS)

Module III - Letter Writing:

Applications, Enquiry, Calling quotations, Tenders, Order and Complaint. (12 HOURS)

Module IV

Precise Writing, Noting and drafting, Technical Description of simple engineering objects and processes (writing), Report writing, precise writing, Note writing, Slogan writing comment, Speech advertising. (12 HOURS)

Module V

Writing Technical reports of the type of observation report, Survey report, Report of trouble, Laboratory Report and Project Report on the subjects of engineering. (Speaking) Vocabulary, Presentations, Demonstrations, Conversation- Telephone media, socializing, cultural events, debates, speech. (12 HOURS)

OUTCOME:

The student understands the below mention:

- Language and skills of communication
- Application linguistic ability
- Letter writing precise writing
- Writing technical reports



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- To be able to represent data in arrays, strings and structure and manipulate them through a program.
- To be able to declare pointers of different types and use them in defining self-referential structures.
- To be able to create, read and write to and from simple text files.

Topics to be covered in the Language laboratory sessions

1. Basic Grammar & Vocabulary (Synonyms /Antonyms, Analogies, sentence completion, correctly spelt words, idioms, proverbs, common errors).
 2. Phonetic symbols and pronunciation.
 3. Listening skills (Including Listening Comprehension)
 4. Reading Skills (Including Reading Comprehension)
 5. Writing Skills (Including structuring resume and cover letter)
 6. Speaking Skills
 7. Body Language
 8. Oral Presentation: Preparation and delivery using audio – visual aids with stress n body language
- And voice modulation (Topic to be selected by the teacher.) Final Assessment Should be based on Assignment, presentation and interview.

BOOKS:

1. Business Correspondence and Report Writing - By Sharma; TMH.
2. Living English Structure – By W.S. Allen; Longmans.
3. English Grammar – Ehrlich, Schaum Series; TMH.
4. Spoken English for India – By R.K. Bansal and IB Harrison Orient Longman.
5. New International Business English – by Joans and Alexander; OUP.
6. Effective Technical Communication – Rizvi; TMH.



SARVEPALLI RADHAKRISHNAN UNIVERSITY, BHOPAL

BE II nd SEMESTER BE 25 ENVIROMENTAL STUDIES

TOTAL -60 HOURS

Module –I Energy-

Sources of Energy : Renewable & Non Renewable, Fossil fuel, Biomass Geothermal, Hydrogen, Solar, Wind, hydal, nuclear sources. (12 HOURS)

Module –II Ecosystem –

Segments of Environment: Atmosphere, hydrosphere, Lithosphere, biosphere. Cycles in Ecosystem – Water, Carbon, Nitrogen. Biodiversity: Threats and conservation, (12 HOURS)

Module –III Air Pollution & Sound Pollution

Air Pollution: Air pollutants, classification, (Primary & secondary Pollutants) Adverse effects of pollutants. Causes of Air pollution chemical, photochemical, Green house effect, ozone layer depletion, acid Rain. Sound Pollution: Causes, controlling measures, measurement of sound pollution (deciblage), Industrial and non – industrial. (12 HOURS)

Module –IV Water Pollution-

Water Pollution: Pollutants in water, adverse effects. Treatment of Domestic & Industrial water effluent.

Soil Pollution – Soil Profile, Pollutants in soil, their adverse effects, controlling measures.

(12 HOURS)

Module –V Society, Ethics & Human values

Impact of waste on society. Solid waste management (Nuclear, Thermal, Plastic, medical, Agriculture, domestic and e-waste). Ethics and moral values, ethical situations, objectives of ethics and its study Preliminary studies regarding Environmental Protection Acts , introduction to value education, self exploration, sanyam & swasthya. (12 HOURS)

BOOKS:

1. Harris, CE, Prichard MS, Rabin's MJ, "Engineering Ethics"; Cengage Pub.
2. RanaSVS; "EssentialsofEcologyand Environment"; PHIPub.
3. Raynold, GW "Ethics in information Technology"; Cengage.
4. Svakumar; Energy Environment & Ethics in society; TMH
5. AK De "Environmental Chemistry"; New Age Int. Publ.



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6. BK Sharma, "Environmental Chemistry"; Goel Publ. House.
7. Bala Krishnamoorthy; "Environmental management"; PHI
8. Gerard Kiely, "Environmental Engineering"; TMH
9. Miller GT JR; living in the Environment Thomson/cengage
10. Cunningham WP and MA; principles of Environment Sc; TMH
11. Pandey, S.N. & Mishra, S.P. Environment & Ecology, 2011, Ane Books , Pvt. Ltd, New Delhi
12. Joseph, B. Environmental Studies, 2009 Tata Mcgraw Hill, Edu India Ltd. New Delhi.
13. Gour R.R, Sangal, R & Bagaria, G.P. , Excel Books, A-45, Naraina Phase-I
New Delhi.-110028



SARVEPALLI RADHAKRISHNAN UNIVERSITY, BHOPAL

BE II nd SEMESTER

BE 26 WORK SHOP/ MANUFACTURING PRACTICE

TOTAL -60 HOURS

OBJECTIVE:

- The student will gain knowledge of the different manufacturing processes which are commonly employed in the industry, to fabricate components using different materials.

Module I

Introduction: Manufacturing Processes and its Classification, Casting, Machining, Plastic deformation and metal forming, Joining Processes, Heat treatment process, Assembly process. Powder Metallurgy: introduction to computers in manufacturing, Black Smithy Shop Use of various smithy tools. Forging operations: Upsetting, Drawing down, Fullering, Swaging, Cutting down, Forge welding, Punching and drafting. Suggested Jobs : Forging of chisel., forging of Screw Driver.

(12 HOURS)

Module II

Carpentry Shop: Timber : Type, Qualities of timber disease, Timber grains, Structure of timber, Timber, Timber seasoning, Timber preservation .Wood Working tools: Wood working machinery, joints & joinery. Various operations of planning using various carpentry planes sawing & marking of various carpentry joints. Suggested Jobs :Name Plate ,Any of the Carpentry joint like mortise or tennon joint. (12 HOURS)

Module III

Fitting Shop: Study and use of Measuring instruments, Engineer steel rule, Surface gauges caliper, Height gauges, feeler gauges, micro meter. Different types of files, File cuts, File grades, Use of surface plate, Surface gauges drilling tapping Fitting operations: Chipping filling, Drilling and tapping.Suggested Jobs :Preparation of job piece by making use of filling, sawing and chipping , drilling and tapping operations. (12 HOURS)

Module IV

Foundry: Pattern Making: Study of Pattern materials, pattern allowances and types of patterns. Core box and core print, Use and care of tools used for making wooden patterns.

Moulding: Properties of good mould & Core sand, Composition of Green , Dry and Loam sand. Methods used to prepare simple green and bench and pit mould dry sand bench mould using single piece and split patterns. (12 HOURS)

Module V

Welding: Study and use of tools used for Brazing, Soldering, Gas & Arc elding. Preparing Lap &



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Butt joints using gas and arc welding methods, Study of TIG & MIG welding processes. Safety precautions. (12HOURS)

OUTCOME:

- Upon completion of the course, students will be able to fabricate components with their own hands.

BOOKS:

1. Bawa HS; Workshop Practice, TMH
2. Rao PN; Manufacturing Technology- Vol.1 & 2, TMH
3. John KC; Mechanical workshop practice; PHI
4. Hazare Choudhary; Workshop Practices -, Vol. I & II
5. Jain. R.K. Production Technology -



SARVEPALLI RADHAKRISHNAN UNIVERSITY, BHOPAL

BE II nd SEMESTER BE 27 Constitution of India

Basic features and fundamental principles

TOTAL -60 HOURS

The Constitution of India is the supreme law of India. Parliament of India cannot make any law which violates the Fundamental Rights enumerated under the Part III of the Constitution. The Parliament of India has been empowered to amend the Constitution under Article 368, however, it cannot use this power to change the "basic structure" of the constitution, which has been ruled and explained by the Supreme Court of India in its historical judgments. The Constitution of India reflects the idea of "Constitutionalism" – a modern and progressive concept historically developed by the thinkers of "liberalism" – an ideology which has been recognized as one of the most popular political ideology and result of historical struggles against arbitrary use of sovereign power by state. The historic revolutions in France, England, America and particularly European Renaissance and Reformation movement have resulted into progressive legal reforms in the form of "constitutionalism" in many countries. The Constitution of India was made by borrowing models and principles from many countries including United Kingdom and America. (20 HOURS)

The Constitution of India is not only a legal document but it also reflects social, political and economic perspectives of the Indian Society. It reflects India's legacy of "diversity". It has been said that Indian constitution reflects ideals of its freedom movement; however, few critics have argued that it does not truly incorporate our own ancient legal heritage and cultural values. No law can be "static" and therefore the Constitution of India has also been amended more than one hundred times. These amendments reflect political, social and economic developments since the year 1950. The Indian judiciary and particularly the Supreme Court of India have played an historic role as the guardian of people. It has been protecting not only basic ideals of the Constitution but also strengthened the same through progressive interpretations of the text of the Constitution. The judicial activism of the Supreme Court of India and its historic contributions has been recognized throughout the world and it gradually made it "as one of the strongest court in the world".

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social and economic developments since the year 1950. The Indian judiciary and particularly the Supreme Court of India has played an historic role as the guardian of people. It has been protecting not only basic ideals of the Constitution but also strengthened the same through progressive interpretations of the text of the Constitution. The judicial activism of the Supreme Court of India

- Course content (20 HOURS)
1. Meaning of the constitution law and constitutionalism
 2. Historical perspective of the Constitution of India
 3. Salient features and characteristics of the Constitution of India
 4. Scheme of the fundamental rights
 5. The scheme of the Fundamental Duties and its legal status
 6. The Directive Principles of State Policy – Its importance and implementation
 7. Federal structure and distribution of legislative and financial powers between the Union and the States
 8. Parliamentary Form of Government in India – The constitution powers and status of the President of India
 9. Amendment of the Constitutional Powers and Procedure
 10. The historical perspectives of the constitutional amendments in India
 11. Emergency Provisions : National Emergency, President Rule, Financial Emergency
 12. Local Self Government – Constitutional Scheme in India
 13. Scheme of the Fundamental Right to Equality
 14. Scheme of the Fundamental Right to certain Freedom under Article 19
 15. Scope of the Right to Life and Personal Liberty under Article 21.

Sarvepalli Radhakrishnan university

(M. P.)

B. Tech II SEMESTER

AU 102 NCC/NSS

Objectives of NCC (National Cadet Corps)

The Student are to be learn the below mentioned

- Develop strong character, discipline comradeship, and leadership qualities.
- Foster a spirit of adventure, sportsmanship and courage
- Promote a secular outlook and selfless service ideas
- Provide basic military training and create a pool of organized youth for national service, potentially serving as a defence reserve

National Cadet Corps (India)

The National Cadet Corps (NCC) is the youth wing of the Indian Armed Forces with its headquarters in New Delhi, India. It is open to school and college students on voluntary basis as a Tri-Services Organisation, comprising the Army, the Navy and Air Force. Cadets are given basic military training in small arms and drill. Officers and cadets have no liability for active military service once they complete their course.

History

The NCC in India was formed in 1948. It can be traced back to the 'University Corps', which was created under the Indian Defence Act 1917, with the objective to make up for a shortage of personnel in the Army. In 1920, when the Indian Territorial Act was passed, the 'University Corps' was replaced by the University Training Corps (UTC). The aim was to raise the status of the UTC and make it more attractive to the youth. UTC Officers and cadets wear Army uniform. It was a significant step towards the 'Indianisation' of the Indian armed forces. It was renamed the UOTC so the National

Cadet Corps can be considered a successor to the University Officers Training Corps (UOTC) which was established by the Government of India in 1942. During World War II, the UOTC never came up to the expectations set by the British. This led to the idea that some better schemes should be formed, which could train more young men in a better way, even during peace. The first Prime Minister of India, Pandit Jawaharlal Nehru presided over the function of raising the first NCC Unit at Delhi on the fourth Sunday of November in 1948. This day is traditionally celebrated as the *NCC Day*. A committee headed by H. N. Kunzru recommended a cadet organization to be established in schools and universities at a national level. The *soldier youth foundation Act* was accepted by the Governor General, and on the 15th of July 1950 the *soldier youth foundation* came into existence.

In 1949, the Girls Division was formed in order to provide equal opportunities to school and college-going girls. The NCC was given an inter-service image in 1950 when the Air Wing was added, followed by the Naval Wing in 1952. In the same year, the NCC curriculum was extended to include community development/social service activities as a part of the NCC syllabus at the behest of Late Pandit Jawaharlal Nehru who took a keen interest in the growth of the NCC. Following the 1962 Sino-Indian War, to meet the requirement of the Nation, NCC training was made compulsory in 1963. This was discontinued in 1968, when the Corps was again made voluntary.

During Indo-Pakistani war of 1965 & Bangladesh-Pakistani war of 1971, NCC cadets were the second line of defence. They organized camps to assist ordnance factories, supplying arms and ammunition to the front and also were used as patrol parties to capture enemy paratroopers. The NCC cadets also worked hand in hand with the Civil defense authorities and actively took part in rescue works and traffic control.

After the 1965 and 1971 wars, the NCC syllabus was revised. Rather than just being a second line of defence, the revised NCC syllabus laid greater stress on developing qualities of leadership and officer like qualities. The military training which the NCC cadets received was reduced and greater importance was given to social service and youth management.

NCC cadets across the country are fully geared up to take part in Civil Defence Mock Drill to be conducted across 244 districts of the country on May 7th, 2025.

Motto and aim

The discussion for the motto of NCC was started in 11th central advisory meeting (CAC) held on 11 August 1978. At that time there were many mottos in mind like "Duty and wisdom"; "Duty, Unity and Discipline"; "Duty and Unity"; "Unity and Discipline". Later, at the 12th CAC meeting on 12 October 1980 they selected and declared "Unity and discipline" as the motto for the NCC. In living up to its motto, the NCC strives to be and is one of the greatest cohesive forces of the nation, bringing together the youth hailing from different parts of the country and molding them into united and disciplined citizens of the nation.

Organisation

The NCC is headed by the Director General (DG), an officer of three-star rank. The DG is assisted by two Additional Director Generals (A and B) of two-star rank (major-general, rear-admiral or air vice-marshal). Five Brigadier level officers and other civil officials also assist him.

The Headquarters is located in Delhi. The organisational structure continues as follows:

- **Directorate** – There are 17 Directorates located in the state capitals headed by an officer of the rank of a Maj Gen from the three Services.
- **Division / Regimental Corps** – There are 3 such Specialised Corps located in Mumbai, Delhi and Bangalore respectively. They are independent of the state directorate and report to the HQ. These divisions form the support function of the regular NCC. Each is headed by a Senior Officer- an equivalent rank of (Lt.) General. Internal Affairs, Administration, Development and Research: Lt. Gen. [SUO] Arvind Shekhar (New Delhi). Recruitment, Training, Media and HR: Lt. Gen. [SUO] Prithvi Pant Negi (Mumbai). Special Forces, Infantry, Gallantry

Committee & Commendations: Lt. Gen. [SUO] Bhav Salimath (Bangalore).

- **Group** – Depending upon the size of the state and growth of NCC in the states, Directorates have up to 14 Group Headquarters under them through which they exercise their command and control of the organisation in the state. Each group is headed by an officer of the rank of Brigadier or equivalent known as Group Commander.
- **Battalion**- Each NCC Group Headquarters control 5–7 units (Bns) commanded by Colonel/Lt.Col or equivalent.
- **Company** – Each Battalion consists of companies which are commanded by the Associate NCC Officer (ANO) of the rank of lieutenant to major.

In all there are 96 Group Headquarters in the country who exercise control over a network of 700 Army wing units (including technical and girls unit), 73 Naval wing units and 64 Air Squadrons. There are two training establishments namely Officers Training School, Kamptee (Nagpur, Maharashtra) and Women Officers Training School, Gwalior. Besides this Vice Chancellor's of various universities across India are conferred with honorary rank of commandant in NCC, to promote and support NCC in their respective University.

Direct or No	Directorates
1	<u>Andhra Pradesh & Telangana</u>
2	<u>Bihar & Jharkhand</u>
3	<u>Delhi</u>
4	<u>Gujarat,Dadra & Nagar Haveli,Daman and Diu</u>
5	<u>Jammu & Kashmir and Ladakh</u>
6	<u>Goa & Karnataka</u>
7	<u>Kerala & Lakshdweep</u>
8	<u>Madhya Pradesh & Chhattisgarh</u>
9	<u>Maharashtra</u>

10	<u>North East Region (Arunachal Pradesh, Assam, Manipur, Meghalaya, Mizoram, Nagaland, Tripura)</u>
11	<u>Odisha</u>
12	<u>Punjab, Haryana, Himachal Pradesh and Chandigarh</u>
13	<u>Rajasthan</u>
14	<u>Tamil Nadu, Puducherry, Andaman and Nicobar Islands</u>
15	<u>Uttarakhand</u>
16	<u>Uttar Pradesh</u>
17	<u>West Bengal & Sikkim</u>

Units

These 17 directorates are divided into 837 units. These units divided into three service groups Army, Naval and Air. Out of those are 700 Army units, 73 Naval and 64 Air units.

Types of Army NCC units and their numbers are given below:

Type of Unit	Number
ARMD SQN	11
ARTY REGT	03
ARTY BTY	20
ENGR REGT	02
ENGR COY	11
SIG REGT	01
SIG COY	13
MED BN	02
MED COY	11
R&V REGT	03
R&V SQN	15
EME BN	02
EME COY	06

Type of Unit	Number
CTR	11
CTC	12
INF BN	369
INF INDEP COY	46
GIRLS BN	97
GIRLS INDEP COY	12

Arms

- HQ: HQ NCC, DTE, Group HQ, BN & COYHQ
- Army: Technical (Engineers, Signals, Medical, EME, CTR),:Non Technical (Infantry, Armoured & ARTY)
- Air: Flying & Technical
- Navy: Unit, Naval Tech., Medical, DAS
- TRG: OTA - Gwalior & OTA – Kamptee

Strength

- Army: Each battalion or unit under the NCC consists of platoons or coy. For senior division boys each platoon consists of 52 cadets and each coy consists 160 cadets. Each BN has 4 to 7 coys so each BN carries around 640 to 1120 cadets. A senior wing girls BN consists of 2 to 7 coy means a total of 320 to 1120 cadets. For junior division boys and junior wing girls each troop has 100 cadets and each BN has at least one troop.
- Navy: For senior division boys each BN or unit consists of 4 to 8 divisions and each division consists of 50 cadets. For senior wing girls similar arrangement exists as of senior division boys. For junior division boys and junior wing girls each BN has a troop of 100 cadets.
- Air: For senior division boys and senior wing girls each unit consists of at least 2 fleets each consists of 100 cadets. So each unit known as Squadron carries around 200 cadets. For junior

wing girls and junior division boys each squadron has a troop consisted of 100 cadets.

However, each unit can have up to 24 troops of senior division boys expanding their strength to 2400 cadets but this is maximum limit.

Personnel

Cadet Ranks

Ranks and Insignias of the NCC								
	Under Officer ranks		SNCO ranks			JNCO ranks		Junior Cadets
Army Wing Insignia								No insignia
Army Wing Ranks	Cadet Senior Under Officer (CSUO)	Cadet Junior Under Officer (CJUO)	Company Sergeant Major (CSM)	Company Quartermaster Sergeant (CQMS)	Sergeant (SGT)	Corporal (CPL)	Lance Corporal (L/CPL)	Cadet (CDT)
Nav y Wing Insignia			No Equivalent				No insignia	No insignia

Navy Wing Ranks	Senior Cadet Captain (SCC)	Cadet Captain (CC)		Petty Officer Cadet (PO CDT)	Leading Cadet (LC)	Naval Cadet I (NC I)	Naval Cadet II (NC II)
Air Force Wing Insignia							No insignia
Air Force Wing Ranks	Cadet Senior Under Officer (CSUO)	Cadet Under Officer (CUO)	Cadet Warrant Officer (CWO)	Sergeant (SGT)	Corporal (CPL)	Leading Flight Cadet (LFC)	Flight Cadet (F/CDT)

JD boys and JW girls are given ranks up to Company Sergeant Major (CSM) in the Army Wing. Only SD boys and SW girls are given ranks above CSM in the Army Wing. In the Air Force and Navy Wings JD boys and JW girls are given ranks up to Cadet Warrant Officer/Petty Officer Cadet only and SD boys and SW girls are given ranks above Cadet Warrant Officer/Petty Officer Cadet.

Regular Officers

The NCC directorates are headed by service officers of the rank of major general and equivalent; group headquarters are headed by service officers of the rank of brigadiers and equivalent. Regular NCC (Army Wing) units are commanded by officers of the rank of Colonel (selection grade); independent NCC Units (Army) are commanded by service officer of the rank of colonel (TS)/lieutenant colonel/major or equivalent. They are responsible for proper training, planning and execution of NCC activities. In addition, each battalion also has an Administration Officer of the rank of Lieutenant Colonel/ Major.

Whole Time Lady Officers

A cadre of whole-time lady officers (WTLO) with cadre strength 110 officers has been sanctioned in 1995. They are to be commissioned partly through departmental channel and partly through UPSC in a phased manner.

Associate NCC Officer

ANO is an important link in the NCC organization between the battalion and the cadets. As a matter of fact, ANO is the feeder node of NCC since they are the one who is in direct contact with the cadets all throughout the year. There are two training establishments namely Officers Training Academy, Kamptee and Officers Training Academy, Gwalior. These two institutions train the school and college teachers selected to head the company/troop. Courses in these institutions range from 21 days to 90 days in duration.

"ANOs are commissioned in NCC and not compare in regular Armed forces".

Associate NCC officers are given the following ranks according to their seniority and their training.

- For colleges (in charge of SD & SW): (NCC Army Wing)
 1. Major
 2. Captain

3. Lieutenant

1. For schools (in charge of JD & JW, equivalent commissioned officer):
2. Chief Officer
3. 1st Officer
4. 2nd Officer
5. 3rd Officer

Uniform

Army NCC cadets wear khaki uniforms, Naval NCC cadets wear the white uniform and Air Force NCC cadets wear grey uniforms. The uniform is compulsory at all meetings and training of the NCC.

Cadets from SD boys Army wing wear khaki full sleeve shirts and trousers. Cadets from JD wear a khaki shirt and khaki trousers. Girl cadets from SW and JW both wear khaki full sleeve shirts and trousers. Cadets from SD boys Naval wing wear white half sleeve shirts and white trousers and JD boys wear half sleeve white shirts and white shorts. Girls from Naval wing SW and JW wear white half sleeve shirts and trousers. Cadets from SD boys Air wing wear light blue half sleeve shirts and trousers & JD cadets wear light blue half sleeve shirts and trousers. Girls from SW and JW wear light blue half sleeve shirts and trousers.

In addition to these uniforms, SW and JW cadets wear white Shalwar kameezes during activities other than when on parade. A rifle green beret is compulsory for all cadets, except Sikh cadets who wear a rifle green turban. For physical training, cadets wear brown canvas shoes, and for drill, they wear black leather shoes called D.M.S (Drill March Shoes). Woolen sweaters are compulsory in cold areas. The colour of these sweaters vary: they are rifle green for the army, black for the navy, and navy blue for the air force. All wear hackles with the beret.

Training

Total training period for SD and SW is 3 years with an extension of 1 year permissible & training period for JD & JW is of 2 years. Every cadet of the Senior or Junior Division has to undergo service training for a period of at least 4 hours per week during the training

year. However, no training is carried out during periods when the college or school through which a cadet is enrolled is closed for a vacation. Every cadet of the Senior and Junior Division has undergone service training for a minimum period of 75% of total hours during the annual college and school session. Every cadet (in case of JD, who has completed one full year of training and is in his second year) attends an annual training camp of 9–10 days, also known as National Combined Annual Training Camp. For SD/SW the duration is usually for up to 30 days. At the end of the camp training the cadets receive a certificate of successful completion.

Certificates & Examination

There are Three Certificates in NCC. Below describes about it from lower value to higher value:-

- Certificate – A : It can be taken by JD/JW cadets of the NCC, during class year 8 and 9 and 10. After passing those classes it can't be obtained. The candidate must have attended a minimum 75% of total training periods laid down in the syllabus for the first and second years of JD/JW NCC (All Wings). The candidate must have attended one Annual Training Camp.
- Certificate – B : It can be taken by SD/SW cadets of the NCC, after class year 10 and those studying for a degree. The candidate must have attended a minimum 75% of total training periods laid down in the syllabus for the first and second years of SD/SW NCC (All Wings). The cadet must have attended one Annual Training Camp/NIC. Cadets who possess Certificate - A will be awarded 10 bonus marks. An air wing cadet must do a minimum 10 Glider launches.
- Certificate – C : Is the highest level certificate for NCC cadets. It can be taken in the third year of training, in the third year of degree course. Those who possess a Certificate - B can take it in the first year after their +2, and in the first year of their degree. The cadet must have attended two Annual Training Camps or one Annual Training Camp and one of the following: RD Camp Delhi, Centrally Organised Camp, Para Training Camp, Attachment Training with service units, National Integration Camp, Youth Exchange Programme, or Foreign Cruise (Navy Wing only).

Grading in Certificates

Three certificates are awarded: 'A' grade, 'B' grade, 'C' grade.

A cadet has to obtain 45% marks in each paper & 50% marks in the aggregate to pass the examination. Grading is based on total marks obtained will be awarded as follows. Grading 'A' – Cadets obtaining 80% marks and above, Grading 'B' – Cadets obtaining 65% marks and above but below 80%, Grading 'C' – Cadets obtaining 50% marks and above but below 65%, Fail – Cadets obtaining less than 45% in any paper or less than 50% in aggregate.

Activities

Republic Day Camp (RDC)

Before the RDC, all group headquarters participate in the IGC (Inter-Group Competition). After the IGC, the selected cadets are trained by the drill instructors of the Indian Army and they are given rigorous military training and drill training daily until the time comes when the cadets have to depart to Delhi to represent their respective states.

- NCC Republic Day Camp is the culmination of all NCC Training activities. RDC is held at Cariappa Parade Ground, Delhi Cantt from 01 to 29 Jan. 2341 Selected NCC Cadets from 17 directorates attend the Camp. The Camp is inaugurated by the Vice President of India and culminates with Prime Minister's Rally on 28 Jan.
- During the camp visit of Raksha Mantri, Cabinet Ministers, Chief Minister of Delhi, Chief of Defence Staff along with various State Ministers/VIPs shall also be organised.
- During the RDC, various competitions are conducted amongst the 17 NCC Directorates to decide the Champion Directorate for award of Prime Minister's Banner. Competitions are keenly contested in various events such as National Integration Awareness presentation, Drill, Line & Flag Area, Cultural Programs i.e. (group song, group dance & ballet), Best Cadet of Senior Division (Boys) and Senior Wing (Girls) in each Service - Army, Navy & Air Force and Best Cadet Boys and Girls each from Junior Wing. Aero modelling and Ship modelling are also conducted during RDC.

- The selections for cadets who will represent India in Youth Exchange Program are also conducted.

The National Cadet Corps maintains two cadet bands: the Boys Band of the NCC and the Girls Band of the NCC. They are commonly formed up during the NCC Republic Day Camp in late January, during which the bands participate in the Republic Day Parade on 26 January and the Prime Minister's Rally on 28 Biggest camps of N.C.C.

Combined Annual Training Camps (CATC) In the CATCs, the boys (senior and junior divisions) and girl cadets (senior and junior wings) of a particular NCC unit participate in the 10-day camp. Classes are conducted as per the given syllabus wherein certain aspects of NCC training are taught once again. The camp acts as a refresher training for the cadets and they are also trained in basic skills of survival and emergencies besides other topics. They are taught certain skills pertaining specifically to their Wing for e.g., a Naval cadet is trained in boat-rowing, oaring, Semaphore, etc. Another activity in the camp is the "dogwatch" wherein two cadets are to stay on sentry duty for two hours at any time given time of the day. Those caught sleeping, especially during late night or early morning shifts, or otherwise missing from duty are severely reprimanded or penalized. Cadets are also introduced to weapons such as a .22 calibre rifle. They are also given tasks of serving food to fellow cadets.

Ek Bharat Shreshtha Bharat Camp (EBSB)

EBSB is to propagate national integration among cadets and society. 'Only the best cadets in drills and marching are sent to represent their states.' This camp is considered for SSLC and higher secondary course (+2) grace marks. These camps are conducted on All-India basis and help bridge the cultural gap among various States of India. In addition, there are six special EBSB campings conducted at Leh, Nagrota (J&K), Chakabama (NER, Nagaland), Srinagar, Lakshadweep and Port Blair.

Advance Leadership Camp (ALC)

Advance Leadership Camp is conducted 6 times in various places throughout India. A cadet must have completed Basic Leadership

Camp in order to qualify for this camp. In this camp the officers give cadets training for SSB screening and entrance.

Army Attachment Camp

These camps are conducted by the NCC in collaboration with Indian Army, as the willing cadets are attached to a specific army battalion undergoing the training period of 10–15 days each. In this camp, the cadets are trained by the instructors of the particular regiment, in the military tactics including day/night warfare & also get familiar with the small arms and heavy weapon systems and are introduced to regimental traditions and customs in infantry regiments and service traditions in other arms and services.

Hiking And Trekking Camps

Adventurous treks and hikes up mountains take place, with expeditions to local mountains and hills.

Thal Sainik Camp (TSC)

The TSC is a 12-day camp conducted in Delhi every year in the late autumn, in which the cadets are selected from all 17 directorates (30+3 SD / JD and SW / JW cadets from each directorate), by the selection procedure conducting 3 pre-TSC camps each of 10–12 days in a week interval. The selected cadets then are sent to the TSC to represent their respective directorates in the following competitions:

- Obstacle course – In which the obstacles include 6-feet wall, zig-zag, double ditch, balancing, 3-feet bar, left bar, right bar, incline, etc. It is done after wearing full tactical gear with rifles.
- Firing – It consists of two types – Shooting (Grouping, Snap-shooting & Application). It is done with a standard .22 caliber rifle at the range of 25 meters & 50 meters.
- Map Reading – which includes working with a compass, service protractor & a map.
- Field craft and battle craft.
- Tent pitching.
- Health and Hygiene.

Vayu Sainik Camp (VSC)

The All India Vayu Sainik Camp is the most prestigious and glorious camp of the NCC air wing. This time AIVSC was held at Air Force Station at Jodhpur in Rajasthan. In-fact, representing at NATIONAL level itself gives you enough power & confidence. The AIVSC is the apex training camp of NCC and is designed to expose the cadets to a strenuous military way of life, in addition to rousing their competitive spirits by pitting them against their peers in a number of aviation-related disciplines such as Microlight Flying, Aero Modelling, Skeet Shooting, .22 Rifle Firing, Drill and written tests of various subjects related to flying. Basically this camp is the culmination of various NCC training activities conducted amongst 16 directorates to decide the champion directorate.

On the first day all the cadets were briefed by the camp commandant regarding camp. Each day there was an activity either it was any competition or any other.

Apart from these events one day was decided to give A visit to Air Force Station, Jodhpur where cadets see and learn how various fighter planes and helicopters works, Luckily cadets also got opportunity to fly in an Indian Air Force Mi-17 helicopter and Pipistrel microlight aircraft and also got chance to visit glorious places of interest in and around Jodhpur.

The camp, in fact, portrays a reflection of 'mini India'. The camp is visited by a number of dignitaries, including DDG and many other army and air force officials.

Nau Sainik Camp (NSC)

NSC is one of the golden camps of NCC, its part of naval wing and conducted under the Indian navy. This centrally organised Naval Camp is conducted annually for selected Naval Wing Cadets. Boat pulling, semaphore, whaler rigging, firing, drill competitions are the main attraction of the camp. The competition is conducted from IGC (inter group completion), PRE NSC 1, 2, 3 and AINSC. It is generally held at Naval Maritime Academy (NAMAC) at Mumbai but started to be held at Karwar from 2014. The coming NSC 2023 is going to be held at INS Shivaji, Lonavala Mumbai from 13 October.

All India Yachting Regetta (AIYR)

This centrally organised Naval Camp is conducted annually for selected Naval Wing Cadets. Yachting (Sailing) is the main attraction of the camp. It is generally held at Naval Base INS Chilika at Odisha.

Rock Climbing Training Camps (RCTC)

Eight rock climbing camps are held each year to expose the cadets to the basics of elementary rock climbing and to inculcate spirit of adventure amongst cadets. Four of these camps are held at Gwalior in Madhya Pradesh and other four camps at Nanyardam near Trivandrum in Kerala.

Naval Wing Activities

Naval wing syllabus is common for both boys and girls. During sea training naval subjects like Seamanship, Navigation, Communication, Gunnery, Damage Control and Ship Safety are taught to cadets. Swimming, Scuba Diving and Wind Surfing are the other interesting activities.

Air Wing Activities

Gliding, Microlite Flying (generally the Zenith STOL CH 701) and attachment training with air force stations and establishments are the main activities. 100 Pipistrel Virus SW 80 on order.

Youth Exchange Programme (YEP)

The aim of YEP is a country-to-country exchange of cadets belonging to NCC or other equivalent government or youth organizations of friendly countries and participation in various activities and appreciation of each other's socio-economic and cultural realities. More than 100 cadets proceed abroad on YEP annually. It is the most prestigious camp in NCC. The cadets currently visit country's namely Singapore, United Kingdom, Mauritius, Vietnam, Maldives, Nepal, Bhutan, Bangladesh, Sri Lanka and Russia. The cadets for YEP are selected annually during the RDC held in Delhi going through various tests and selection criteria.

Cadets from foreign nations also visit India in YEP every year during RDC and Desert Safari held in Rajasthan DTE. Around 25 countries participate in PM Rally during RDC.

Overseas Deployment

A select few cadets from the Senior Division (Navy) are attached to the 1st Training Squadron of the Indian Navy, for a period of 30–45 days, wherein they are trained in Naval Subjects and Practical Seamanship as well as travel to friendly foreign nations on Goodwill Missions. A total of 10-20 cadets are selected to represent the National Cadet Corps, for this camp.

The Director-Generals of the NCC held the rank of major-general from 1951 until 1983, when the appointment was upgraded to the rank of Lieutenant General.

Sarvepalli Radhakrishnan university

(M. P.)

B. Tech II SEMESTER

AU 102 NCC/NSS

Objectives of NSS (National Service Scheme)

- Promote national unity and patriotism
- Develop the personality and character of student youth through voluntary community service
- Develop a sense of social and civic responsibility and democratic attitudes
- Gain skill in mobilizing community participation and meeting emergencies/ disasters
- Practice national integration and social harmony
- Offer opportunities for leadership empathy, and broadening life skills, inspired by Mahatma Gandhi's principles

National Service Scheme

Introduction

In India, the idea of involving students in the task of national service dates back to the times of Mahatma Gandhi, the father of the nation. The central theme which he tried to impress upon his student audience time and again, was that they should always keep before them, their social responsibility. The first duty of the students should be, not to treat their period of study as one of the opportunities for indulgence in intellectual luxury, but for preparing themselves for final dedication in the service of those who provided the sinews of the nation with the national goods & services so essential to society. Advising them to form a living contact with the community in whose midst their institution is located, he suggested that instead of undertaking academic research about economic and social disability, the students should do "something positive so that the life of the villagers might be raised to a higher material and moral level".

The post-independence era was marked by an urge for introducing social service for students, both as a measure of educational reform and as a means to improve the quality of educated manpower. The University Grants Commission headed by Dr. Radhakrishnan recommended introduction of national service in the academic institutions on a voluntary basis with a view to developing healthy contacts between the students and teachers on the one hand and establishing a constructive linkage between the campus and the community on the other hand.

The idea was again considered by the Central Advisory Board of Education (CABE) at its meeting held in January, 1950. After examining the various aspects of the matter and in the light of experience of other countries in this field, the Board recommended that students should devote some time to manual work on a voluntary basis and that the teachers should also associate with them in such work. In the draft First Five Year Plan adopted by the Government of India in 1952, the need for social and labour service for students for one year was further stressed. Consequent upon this, labour and social service camps, campus work projects, village apprenticeship scheme etc. were put into operation by various educational institutions. In 1958, the then Prime Minister Pandit Jawaharlal Nehru in his letter to the

Chief-Ministers, mooted the idea of having social service as a prerequisite for graduation. He further directed the Ministry of Education to formulate a suitable scheme for introduction of national service into the academic institutions.

In 1959, a draft outline of the scheme was placed before the Education Minister's Conference. The Conference was unanimous about the urgent need for trying out a workable scheme for national service. In view of the fact that education as it was imparted in schools and colleges, left something to be desired and it was necessary to supplement it with programmes which would arouse interest the social and economic reconstruction of the country. It was viewed that if the objectives of the scheme were to be realized, it was essential to integrate social service with the educational process as early as possible. The Conference suggested the appointment of a committee to work out details of the proposed pilot project. In pursuance of these recommendations, a National Service Committee was appointed under the Chairmanship of Dr. C.D. Deshmukh on August 28, 1959 to make concrete suggestions in this direction. The committee recommended that national service for a period of nine months to a year may be made compulsory for all students completing high school education and intending to enroll themselves in a college or a university. The scheme was to include some military training, social service, manual labour and general education. The recommendations of the Committee could not be accepted because of its financial implications and difficulties in implementation.

In 1960, at the instance of the Government of India, Prof. K.G. Saiyidain studied national service by students implemented in several countries of the world and submitted his report under the title "National Service for the Youth" to the Government with a number of recommendations as to what could be done in India to develop a feasible scheme of social service by students. It was also recommended that social service camps should be open to students as well as non-students within the prescribed age group for better inter-relationship.

The Education Commission headed by Dr. D.S. Kothari (1964-66) recommended that students at all stages of education should be associated with some form of social service. This was taken into account by the State Education Minister during their conference in April 1967 and they recommended that at the university stage, students could be permitted to join the National Cadet Corps (NCC) which was already in existence on a voluntary basis and an alternative to this could be offered to them in the form of a new programme called the National Service Scheme (NSS). Promising sportsmen, however, should be exempted from both and allowed to join another scheme called the National Sports Organisation (NSO), in view of the need to give priority to the development of sports and athletics.

The Vice-Chancellors' Conference in September, 1969 welcomed this recommendation and suggested that a special committee of Vice Chancellors could be set up to examine this question in detail. In the statement of national policy on education of the Government of India, it was laid down that work experience and national service should be an integral part of education. In May, 1969, a conference of the students' representatives of the universities and institutions of higher learning convened by the Ministry of Education and the University Grants Commission also unanimously declared 'that national service could be a powerful instrument for national integration. It could be used to introduce urban students to rural life. Projects of permanent value could also be undertaken as a symbol of the contribution of the student community to the progress and upliftment of the nation.

The details were soon worked out and the Planning Commission sanctioned an outlay budget of Rs. 5 crores for National Service Scheme (NSS) during the Fourth Five Year Plan. It was stipulated that the NSS programme should be started as a pilot project in selected institutions and universities.

On September 24, 1969, the then Union Education Minister Dr. V.K.R.V. Rao, launched the NSS programme in 37 universities covering all States and simultaneously requested the Chief Ministers of States for their cooperation and help. It was appropriate that the programme was started during the Gandhi Centenary Year as it was Gandhi Ji who inspired the Indian youth to participate in the movement for India's independence and the social upliftment of the downtrodden masses of our nation.

The cardinal principle of the programme is that it is organised by the students themselves and both students and teachers through their combined participation in social service, get a sense of involvement in the tasks of national development. Besides, the students, particularly, obtain work experience which might help them to find avenues of self-employment or employment in any organisation at the end of their university career. The initial financial arrangements provided for an

expenditure of Rs. 120/- per NSS student per annum to be shared by the Central and the State Governments in the ratio of 7:5 i.e. the Central Government spending Rs. 70/- and State Governments Rs. 50/- respectively per NSS student per year. An amount of Rs. 120/- per NSS student per annum on programmes to be shared by the Central and State Governments in the ratio of 7:5 (i.e. Rs. 70/- per student by the central government and Rs. 50/- per student by the State Governments). Now, the NSS is a Central Sector Scheme of Government of India and to run its activities & Programmes in all the States and Union Territories are fully funded by the Central Government. Keeping the inflation in view, the budget for Special Camping and Regular Activities is as follows:

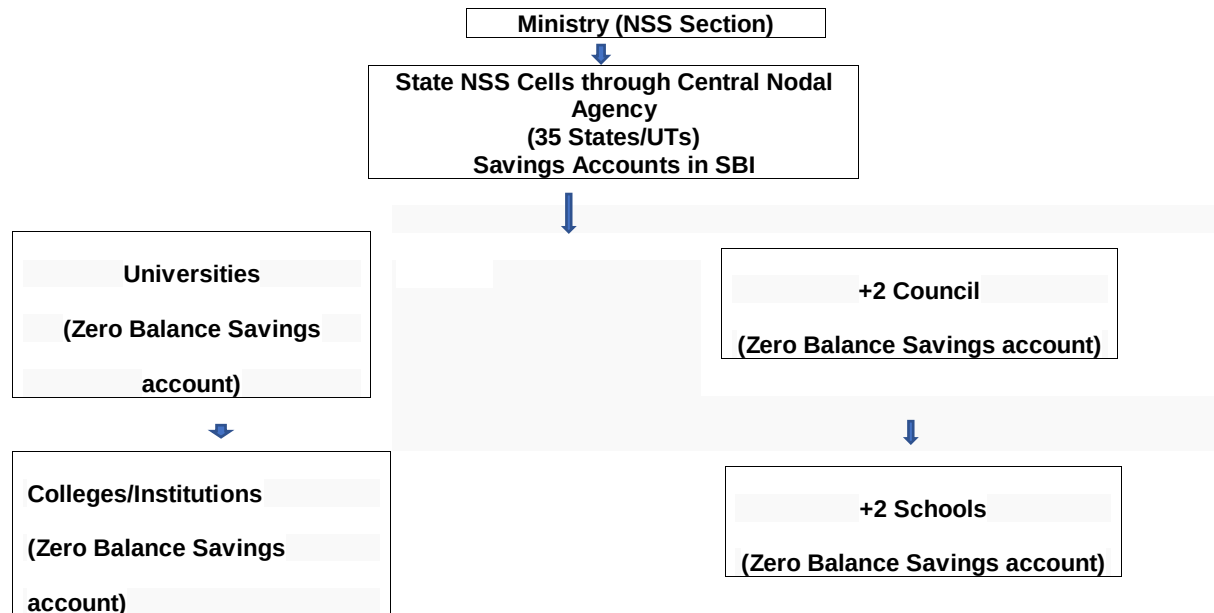
- Regular Activities: Rs.400/- per volunteer, per annum.
- Special Camping Programme: Rs.700/- per volunteer, for two consecutive years.
 - The scheme now extends to all the states/Union Territories and universities in the country and covers Councils of +2 level in the country also in many states. Students, teachers, guardians, persons in authority in government, universities and colleges/schools and the people in general now realize the need and significance of NSS. It has aroused among the student youth an awareness of the realities of life, a better understanding and appreciation of the problems of the people. NSS is, thus, a concrete attempt in making campus relevant to the needs of the community. There are several instances of excellent work and exemplary conduct of NSS units which have earned them appreciation respect and confidence of the people. The special camping programmes organised under the themes of 'Youth Against Famine (1973)', 'Youth Against Dirt & Disease (1974-75)', 'Youth for Eco-Development' and 'Youth for Rural Reconstruction' 'Youth for National Development and Youth for Literacy (1985-93)' 'Youth for National Integration and Communal Harmony (1993-95)' have resulted in gains both to the community as well as to the students. The theme for the year 1995-96 onwards for Special Camping is 'Youth for Sustainable Development with focus on Watershed Management and Water land Development'. Themes have been selected in accordance with national priorities. Also, from 1991-92 onwards NSS has launched a nationwide campaign on AIDS Awareness called "Universities Talk AIDS" (UTA) which has earned international attention and appreciation.
 - Community service rendered by university and +2 level students has covered several aspects like adoption of villages & Urban slums for intensive development work, carrying out the medico-social surveys, setting up of medical centres, programmes of mass immunization, sanitation drives, adult education programmes for the weaker sections of the community, blood donation, helping patients in hospitals, helping inmates of orphanages and the physically handicapped etc. NSS volunteers rendered commendable relief work during natural calamities/emergencies such as cyclones, floods, famine, earthquake, tsunamis etc from time to time all over the country. The NSS students have also done useful work in organising campaigns for eradication of social evils, and popularization of the nationally accepted objectives like nationalism, democracy, secularism, social harmony and development of scientific temper.

The main objectives of National Service Scheme (NSS) are:

- i. understands the community in which they work
- ii. understand themselves in relation to their community
- iii. identify the needs and problems of the community and involve them in problem-solving
- iv. develop among themselves a sense of social and civic responsibility
- v. utilises their knowledge in finding practical solutions to individual and community problems
- vi. develop competence required for group-living and sharing of responsibilities
- vii. gain skills in mobilising community participation
- viii. acquire leadership qualities and democratic attitudes

- ix. develop capacity to meet emergencies and natural disasters and
- x. practise national integration and social harmony

NSS Fund Flow Schematic



A) Central Level – Ministry (NSS Section)

- The Ministry is the main authority that allocates and manages NSS funds at the national level.

B) State Level – State NSS Cells via Central Nodal Agency

- Funds are transferred from the Ministry to State NSS Cells.
- There are 35 States/UTs, and each operates through a Central Nodal Agency.
- These agencies maintain Savings Accounts in SBI (State Bank of India) for fund transactions.

C) Disbursement to Implementing Bodies

- From the State NSS Cells, funds flow down to the respective institutions via zero balance savings accounts (accounts that do not require a minimum balance).
- There are two main parallel branches:
 - Higher Education Institutions
 - Universities → Funds go to their Zero Balance Savings Account.
 - Then, from universities to Colleges/Institutions, also having Zero Balance Savings Accounts.

Regular Activities

NSS volunteers generally work in villages, slums and voluntary agencies to complete 120 hours of regular activities during an academic year. As per the fundamental principles of National Service Scheme, a volunteer is expected to remain in constant touch with the community. Hence, it is of vital importance that a particular village/slum is selected for implementation of NSS programmes. As the NSS volunteer is to live with the members of the community during the 7 days Special Camping Programme and learn from their experience during his/her tenure in NSS, the village/slum should be carefully selected for adoption by NSS unit.

1. Adoption of Villages

1.1 Adoption of a village and area is a very meaningful programme in NSS. It is far better to concentrate attention on one village and take up the task for development perspective, than to fritter away energy in many locations involving too many activities which may not be completed at all or where the follow up action may not be possible. From this point of view, village adoption programme should ensure continuity of work vis-a-vis sustained action, evaluation and follow up work.

Contacting Village/Area Leaders

1.2 As a first step in this programme, it is necessary to establish contact with more than one village which would help to select a village where 'Leadership' is well established. In other words, selecting a village with proper leadership is very important as the sustained follow up action and evaluation is ensured in such places. To start with, the NSS unit can take the help of the Block Authorities, District Panchayat Officer, District Tribal Welfare Officer, District Medical Officer, Extension Officer of Agriculture, Irrigation and Education Departments for the selection of the village. It is to be noted that the selected villages should be within a short distance from the college so that constant contact can easily be made.

Survey of the Village/Area

1.3 Before drawing up the plan of action, it is absolutely necessary to conduct a comprehensive survey of few villages situated at a short distance from the college. The assistance from the teachers and students of agriculture, economics, commerce, geography, statistics, home science, social work, medicine, psychology and education etc. have to be sought for the purpose. Conducting socio-economic survey can be an interesting field activity which has direct bearing on the curriculum of economics, commerce, statistics, psychology, health education etc. The report of such a survey will provide up-to-date information about the problems and potentials of the village and help in programme planning for village development. The applied field work will help the students to increase their analytical ability and deepen their thinking. Further, this will help them to identify the problems which have been left unnoticed. The survey work can also be accomplished with the help of PRA exercises (Participatory Rural Appraisal)

Identification of Problem(s)

1.4 It is on the basis of this need assessment that projects/programmes are to be formulated. The programme officers should use their discretion and should identify the projects which can be completed by seeking assistance from the communities/other agencies.

1.5 The aim of adoption of village or area is to give new ideas of development to the villagers which would improve their living conditions. Once the trust of the communities is won, they start cooperating with the NSS volunteers and approach them for solution of their problems. One of the important services that can be rendered by NSS volunteers is disseminating information about the latest developments in agriculture, watershed management, wastelands development, non-conventional energy, low cost housing, sanitation, nutrition and personal hygiene, schemes for skill development, income generation, government schemes such as Swachh Bharat, Ayushman Bharat, Accessible India, Digital India, Beti Bachao and Beti Padhao, Environment and Energy Conservation and Education, legal aid, consumer protection and allied field.

1.6 The Programme Officers (PO) should motivate the communities to involve themselves with NSS for the community development work undertaken by the NSS unit. Further he or she will have

to seek the help of various government departments and agencies for technical advice and financial assistance. Therefore, he/she must establish good rapport with the government officials and development agencies. For this, it is better if the administration is taken into confidence by prior consultations.

Completion of Projects

1.7 As already stated, the Programme Officer must select the projects very carefully as the image of NSS depends upon the successful completion of such projects. Successful completion of the projects can win appreciation and credit of the community'

Evaluation of Project

1.8 Every project should be evaluated after its completion by involving members of the community, Government officials and Panchayat officials. The NSS unit should learn from the lapses in the execution of the project and plan for the next project keeping in view the bottlenecks and constraints faced by them during the earlier project.

2. Adoption of Slums

Most of the colleges and universities are usually located in the urban areas. Due to long distance between the college campuses and the villages, the visits to the adopted villages by the NSS volunteers may become expensive and time consuming. In view of this, it is desirable to adopt slum especially by colleges located in urban areas.

Survey of the Slum

2.1 For adoption of a slum, there should be composite survey team consisting of students drawn from faculties like, Arts, Science, Engineering, Medicine, Home Science and Social Work etc. The selected areas should be compact and should be easily accessible for students. Areas with acute political conflicts may be avoided.

2.2 The issues pertaining to the identification of problems, project planning, interaction and coordination with the various departmental agencies, execution and completion of projects shall be undertaken on the same basis/lines as discussed earlier in the part 'Adoption of Villages'. The progress of projects should also be reviewed frequently.

Services in Slums

2.3 The slum, tenements, Jhuggis and Jhoupris can be adopted by the NSS units with the aim of slum improvement. Under this, activities like providing water, water logging, sanitation, electricity, drainage, health and welfare services, life and living conditions, can be undertaken.

NSS volunteers for Slum Work

2.4 Taking into account the living conditions & status of slum dwellers of the slums, only highly motivated, adaptable, mature and skilled students should be selected for slum development.

2.5 Following are the tasks which the students can undertake in slum areas:-

(a) As Community Investigators: They can prepare brief community profile on various slums in the city or town covering different amenities, services, and living conditions etc.

(b) As Community Workers: They can identify local leaders and in cooperation with them discuss local problems on which cooperative action can be initiated.

(c) As Programme Aides: Students can help the local communities' in launching a number of programmes like setting up a free milk distribution centre, sanitation drive, recreation, adult lad primary education, health projects like immunization. first aid centre, child care, nutrition classes, and free legal aid centres etc. They can also help in forming youth clubs, children's groups, mahila mandals etc.

(d) As Community Organizers: NSS students, after establishing rapport with the slum dwellers, can form community association to tackle local problems on a group basis with reliance on local resources, self help and mutual aid and with some minimum external assistance.

2.6 Some suggestions for Selection of slums

(i) There should be a socio-economic survey of the slum by a team of volunteers drawn from different faculties;

(ii) The selected area should be compact. There should not be more than 300 residents in an adopted slum;

(iii) The community people should be receptive to the ideas of improving their living standard. They should also be ready to coordinate and involve in the projects undertaken by the NSS for their upliftment;

(iv) The areas where political conflicts are likely to arise should be avoided by the NSS units;

(v) The area should be easily accessible to the NSS volunteers to undertake frequent visits to slums;

(vi) The working in the slum needs commitment and hard work. Only sensitive and highly motivated NSS volunteers can find easy to serve in slum areas.

3. Coordination with Voluntary Organizations

It may be noted that the NSS unit has no financial resource to implement any programme in the adopted villages or slum on its own. Therefore, a successful unit has to closely coordinate with the government agencies and voluntary organisations working in this field.

3.1 After identification of the needs of the community and the selection of projects, the programme officer should look for the Government agency or a voluntary organisation who can assist in the completion of a particular project. The different departments of the Government like forest, agriculture, adult education, health, child and family welfare, can render very useful assistance to the project pertaining to community work. Voluntary organisations can also help in forming public opinion in favour of NSS projects. Adult education samitis, Nasha Bandi Boards, Yuva Mandals and Mahila Mandals can provide additional assistance to the NSS units. Similarly voluntary land statutory welfare agencies such as Community Centres, Residential Institutions for children/women, the Aged and Institutions for physically handicapped and disabled can provide a wide scope of the choice of service to the NSS volunteers. NSS volunteers can be placed with these agencies considering the inherent aptitudes and inclinations of NSS volunteers. NSS volunteers must be told to develop a sense of belonging and respect for the people with whom they are working. Working in close collaboration with these agencies will help the NSS volunteers in understanding the problems of a vulnerable section of the society. The work opportunities in the welfare institutions may be enumerated as under-

(i) Adoption of welfare institutions and helping the inmates and staff by arranging outings, fund collection drives, reading and writing letters for those who are unable to do so;

(ii) propagation of the message of Swachhta, Digital Literacy, renewable energy etc.

(iii) working for improvement of physical environment;

(iv) programme of non-formal education and general literacy classes;

(v) organisation of economic development activities;

(vi) establishment of hobby centres and

(vii) assisting in the rehabilitation work of the disabled, destitute etc.

In addition, NSS units and welfare agencies can take up joint community development projects and other programmes of community welfare and awareness depending on the local needs.

3.2 The NSS Programme Officers should plan activities in the adopted village or slum in such a way that the leisure time of the NSS volunteers can be utilized in the service of adopted village or slum. Week-end visits to the adopted areas provide suitable opportunities to live with the community and know their problems and make an earnest effort to do something for them. Similarly efforts should be made to follow up the work done in the areas earlier. Such sustained efforts will flower into friendship between the NSS unit and the community. These activities can be arranged through one day camps and frequent visits under regular activities.

Special Camping Programme

Special Camping forms an integral part of National Service Scheme. It has special appeal to the youth as it provides unique opportunities to the students for group living, collective experience sharing and constant interaction with community.

1. Special campings are organised generally on various developmental issues of national importance. In the past the themes of the Special Camping Programmes have been 'Youth

Against Famine', 'Youth Against Dirt and Disease', 'Youth for Rural Reconstruction', 'Youth for Eco-Development' and 'Youth for Mass Literacy', 'Youth for National Integration & Social Harmony'. 'Youth for Sustainable Development with special focus on Watershed Management and Wasteland Development. Every year 50 percent of the volunteers of each NSS unit are expected to participate in special camps which is of seven days duration.

Contributions of Special Camping Programme

1.1 Concerted efforts have to be made for a number of years for reconstruction activities in rural areas and urban slums for improving the living conditions of economically and socially weaker sections of the community. For this, the universities colleges land +2 institutions having NSS have a special role to play in collaboration with other Departments and local authorities engaged in Development work. They adopt a village or group of villages/urban slums for intensive social development, where special camps are to be organised by them year after year to create tangible and durable community assets.

Objectives of the Special Camping programme

1.2 The primary objectives of the special camping programmes are:-

- (i) Making education more relevant to the present situation to meet the felt needs of the communities and supplement the education of university/college/school students by bringing them face to face with the community situation.
- (ii) To provide opportunities to NSS Volunteers to play their due roles in the implementation of various development "programmes by planning and executing development projects, which not only help in creating durable community assets in rural areas and slums but also result in improvement of the condition of weaker sections of the communities.
- (iii) Encouraging the students and non-students youth to work along with the adults in rural areas, thereby developing their character, social consciousness and commitment, discipline and healthy and helpful attitudes towards the community:
- (iv) Building up potential youth leaders by exploring the latent potential among the campers, both students as well as local youth (rural and urban), with a view to involve them more intimately in development projects for longer periods. The local leadership generated during the camps would also be useful in ensuring proper maintenance of the assets created as a result of the camps.
- (v) Emphasizing the dignity of labour and self-help and the need for combining physical work with intellectual pursuits, and
- (vi) Encouraging youth to participate enthusiastically in the process of national development, and promote national integration through democratic living and cooperative action.

Suggestive list of activities during Regular as well as Special Camping

1.3 The aim of the Regular and special Camping Programme is to bring youth face to face with the community and make efforts to improve their life. The NSS volunteers are to devote about 80 hours in Regular Activities for the development of the adopted village. Special Camping has been conceived as an opportunity to live with that community for 7 days, and experience the conditions and problems of the people. The NSS volunteers need to be inspired to take initiatives for the improvement of their condition. Although the focus of the Special Camps change periodically and regular programmes are organized in response to the community needs at the micro-level, some broad areas of activities are enumerated below:-

(a) Environment Enrichment and Conservation: Where as there would be a main theme for the special camping Programme, activities aimed at environment - enrichment would be organized under the sub-theme of" Youth for Better Environment". The activities under this sub-theme would inter-alia, include:

- (i) Plantation of trees, their preservation and upkeep (each NSS unit should plant and protect at least 1000 saplings);
- (ii) Creation of NSS parks/gardens.
- (iii) Construction & maintenance of village streets, drains, etc. so as to keep the environment clean;

- (iv) Construction of sanitary latrines etc.
- (v) Cleaning of village ponds and wells;
- (vi) Popularization and construction of Gobar Gas Plants, use of non-conventional energy;
- (vii) Environmental sanitation and disposal of garbage & composting;
- (viii) Prevention of soil erosion, and work for soil conservation,
- (ix) Watershed management and wasteland development
- (x) Preservation and upkeep of monuments, and creation of consciousness about the preservation of cultural heritage among the community.

(b) Health, Family Welfare and Nutrition Programme:

- (i) Programme of mass immunization;
- (ii) Working with people in nutrition programmes with the help of Home Science and medical college students;
- (iii) Provision of safe and clean drinking water;
- (iv) Integrated child development programmes;
- (v) Health education, AIDS Awareness and preliminary health care.
- (vi) Population education and family welfare programme;
- (vii) Life style education centres and counseling centres.

(c) Programmes aimed at creating an awareness for improvement of the status of women:

They may, inter-alia, include:

- (i) Programmes of educating people and making them aware of women's rights both constitutional and legal;
- (ii) Creating consciousness among women that they too contributed to economic and social well-being of the community;
- (iii) Creating awareness among women that there is no occupation or vocation which is not open to them provided they acquire the requisite skills; and
- (iv) Imparting training to women in sewing, embroidery, knitting and other skills wherever possible.
- (v) Beti Bachao Beti Padhao.

(d) Social Service Programmes:

Depending on the local needs and priorities, the following activities/programmes may be undertaken:-

- (i) Work in hospitals, for example, serving as ward visitors to cheer the patients, help the patients, arranging occupational or hobby activities for long term patients, guidance service for out-door-patients including guiding visitors about hospital's procedures, letter writing and reading for the patients admitted in the hospital; follow up of patients discharged from the hospital by making home visits and places of work, assistance in running dispensaries etc.
- (ii) Work with the organisations of child welfare;
- (iii) Work in institutions meant for physically and mentally handicapped;
- (iv) Organising blood donation, eye pledge programmes;
- (v) Work in Cheshire homes, orphanages, homes for the aged etc.;
- (vi) Work in welfare organisations of women;
- (vii) Prevention of slums through social education and community action;

(e) Production Oriented Programmes:

- (i) Working with people and explaining and teaching improved agricultural practices;
- (ii) Rodent control and pest control practices;
- (iii) Weed control;
- (iv) Soil-testing, soil health care and soil conservation;
- (v) Assistance in repair of agriculture machinery;
- (vi) Work for the promotion and strengthening of cooperative societies in villages;

- (vii) Assistance and guidance in poultry farming, animal husbandry, care of animal health etc.;
- (viii) Popularization of small savings and
- (ix) Assistance in procuring bank loans

(f) Relief & Rehabilitation work during Natural Calamities:

These programmes would enable the students to understand and share the agonies of the people affected in the wake of natural calamities like cyclone, flood, earthquakes, etc. The main emphasis should be on their participation in programmes, and working with the people to overcome their handicaps, and assisting the local authorities in relief and rehabilitation work in the wake of natural calamities. The NSS students can be involved in:-

- (i) assisting the authorities in distribution of rations, medicine, clothes etc.;
- (ii) assisting the health authorities in inoculation and immunization, supply of medicine etc.;
- (iii) working with the local people in reconstruction of their huts, cleaning of wells, building roads etc.;
- (iv) assisting and working with local authorities in relief and rescue operation;
- (v) collection of clothes and other materials, and sending the same to the affected areas;

(g) Education and Recreations:

Activities in this field could include:

- (i) adult education (short-duration programmes);
- (ii) pre-school education programmes;
- (iii) programmes of continuing education of school drop outs, remedial coaching of students from weaker sections;
- (iv) work in creches ;
- (v) participatory cultural and recreation programmes for the community including the use of mass media for instruction and recreation, programmes of community singing, dancing etc.;
- (vi) organisation of youth clubs, rural land indigenous sports in collaboration with Nehru Yuva Kendras;
- (vii) programmes including discussions on eradications of social evils like communalism, castism, regionalism, untouchability, drug abuse etc.;
- (viii) non-formal education for rural youth and
- (ix) legal literacy, consumer awareness.
- (X) Swacch Bharat Mission
- (XI) Digital awareness,
- (XII) Voter awareness.

1.4 The above is only an illustrative list of the type of activities that can be undertaken, Under the programme it would be open to each NSS Unit to undertake one of these programmes or any other activity which may seem desirable to them according to local needs, The NSS Unit should aim at the integrated development of the area selected for its operation which could be a village or a slum. It has also to be ensured that at least a part of the programme does involve manual work.